

Field Demonstrations of Operational Strain Modal Analysis of a Historic Suspension Bridge Using Installed Telecom Fibres

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Abstract *We demonstrate operational strain modal analysis of the Menai suspension bridge using installed telecom fibre and distributed acoustic sensing. Using enhanced frequency-domain decomposition, we extract key structural health indicators, including natural frequencies, mode shapes, and damping ratios, providing a cost-effective, continuous, and robust SHM solution. ©2026 The Author(s)*

Introduction

Historic suspension bridges are vital infrastructure assets whose durability is increasingly affected by ageing, rising traffic, and environmental factors. Maintaining their structural health requires ongoing and dependable monitoring [1]. Among various methods, modal analysis provides critical information about how the structure behaves by examining key indicators of structural conditions, including modal frequencies, shapes, and damping ratios.

Traditional structural monitoring relies on Experimental Modal Analysis (EMA), which requires controlled excitations and is impractical for large-scale bridges. Operational Modal Analysis (OMA) overcomes this limitation by using ambient excitations such as wind and traffic, enabling monitoring under normal operating conditions [2]. However, the OMA approaches typically depend on discrete sensors, such as accelerometers or strain gauges, resulting in sparse spatial coverage and limited capability for precise spatial localisation [3]. Distributed Acoustic Sensing (DAS) has emerged as a powerful alternative by transforming optical fibres into dense arrays of strain sensors, enabling spatially continuous measurements along the entire structure [4].

However, recent applications of DAS in bridge monitoring have predominantly relied on dedicated fibre installations, while the use of existing telecom fibre infrastructures remains limited. Moreover, current studies are mainly focused on short- to medium-span structures and vibration detection [5, 6]. The use of DAS for modal analysis, particularly strain-based operational modal analysis on long-span suspension bridges, remains largely unexplored.

To address this limitation, this paper demonstrates Operational Strain Modal Analysis (OSMA) for the 200-year-old Menai suspension

bridge by analysing DAS data collected through a pre-existing telecom fibre installed along the bridge. To the best of our knowledge, this is the first field demonstration of DAS-based OSMA for a suspension-type bridge. Unlike other DAS-based approaches that derive displacement from strain measurements through spatial integration [7], the proposed method directly exploits spatially continuous strain data for modal identification. This not only improves measurement sensitivity for mode shape and spatial resolution but also establishes a new paradigm for structural health monitoring. While traditional OMA primarily serves as a global damage detection tool, the strain-based modal representation provides a physically meaningful pathway toward spatially resolved damage localisation, offering a foundation for future high-precision structural diagnostics [8].

We apply the Enhanced Frequency-Domain Decomposition (EFDD) method to capture DAS-phase data to identify natural frequencies, strain mode shapes, and damping ratios. The modal consistency is evaluated using the Modal Assurance Criterion (MAC) across different time periods. The dominant natural frequencies for the first two modes are estimated as 0.51 Hz and 0.73 Hz. A MAC index over 0.92 is observed across different time periods, thereby justifying the temporal consistency and robustness of the proposed technique.

Setup and Methodology

The field experiment to monitor the historic Menai suspension bridge was conducted using a commercial DAS system (Helios interrogator) connected to an installed telecom fibre along the bridge. As illustrated in Figure 1(a), the field fibre, connected in a 57-km loop, is installed in the Bangor and Anglesey area of North Wales, running across two bridges: the Menai suspension bridge and the Britannia bridge. The

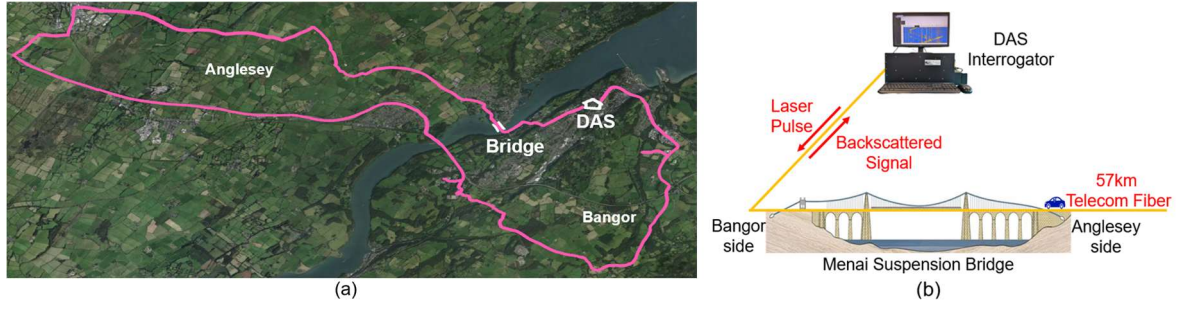


Fig. 1: Experimental setup. (a) Field deployment showing the 57-km telecom fibre route (red line) linking the DAS interrogator (DAS) and the Menai Suspension Bridge (Bridge). (b) Schematic of the DAS measurement principle.

DAS was connected at one end of the fibre, as shown in Fig. 1(b), on the Bangor side, approximately 4.7 km from the DAS equipment.

The DAS, configured with a pulse width of 100 ns and a pulse repetition rate of 600 Hz, recorded optical phase variations of the backscattered signal at a sampling rate of 150 MS/s. To evaluate structural dynamic response, 17 sensing channels (*i.e.*, 17 measurement points) across the main span were extracted, providing ultra-dense spatial coverage for high-resolution monitoring.

Continuous data acquisition was performed under broadband ambient excitations, primarily driven by traffic loads and wind. The DAS system measures optical phase variations $\Delta\phi(z, t)$ along the fibre, which are directly proportional to the longitudinal strain $\epsilon(z, t)$ experienced by the fibre as [9]:

$$\Delta\phi(z, t) = \frac{4\pi n \epsilon}{\lambda} \cdot L \cdot \epsilon(z, t) \quad (1)$$

where n is the refractive index, ϵ is the photo-elastic scaling factor, λ is the optical wavelength, and L is the gauge length. While traditional bridge monitoring often attempts to reconstruct displacement via spatial integration of strain, such operations are highly sensitive to noise accumulation and require precisely defined boundary conditions [7]. For long-span suspension bridges, complex cable-deck interactions and spatially varying stiffness make accurate displacement reconstruction technically challenging and error-prone. Consequently, this study directly utilises spatially continuous strain data for modal identification, which offers superior sensitivity to localise structural anomalies and stress concentrations.

Prior to modal identification, the raw DAS-phase signal were pre-processed to enhance data quality and consistency. Linear detrending and channel-wise normalisation were applied to remove low-frequency baseline drift and ensure uniform amplitude scaling across the span. The pre-processed data were segmented into 10-minute non-overlapping windows, and Power Spectral Density (PSD) matrices were computed

for each window using Welch's method to study the spectral characteristics, focusing on the low frequency band relevant dynamic response.

Then, the modal parameters were identified using the EFDD method [10]. The core of this approach involves the Singular Value Decomposition (SVD) of the estimated PSD matrix $\hat{G}_{yy}(j\omega_i)$ calculated from raw vibration data at each discrete frequency ω_i :

$$\hat{G}_{yy}(j\omega_i) = U_i S_i U_i^H \quad (2)$$

where S_i is a diagonal matrix containing the scalar singular values and $U_i = [u_{i1}, u_{i2}, \dots, u_{in}]$ is a unitary matrix that contains the singular vectors. Natural frequencies are identified by detecting the frequencies at which the peaks occur at the first singular value. Mode shape is estimated as the first singular vector, u_{i1} , at a particular natural frequency. After that, for each identified peak, the isolated auto-spectral density within a narrow frequency band is transformed to the time domain via IFFT to obtain the Auto-Correlation Function (ACF). The damping ratio ζ is then estimated using the logarithmic decrement δ [11]:

$$\delta = \ln\left(\frac{x_n}{x_{n+1}}\right) = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}} \quad (3)$$

where x_n and x_{n+1} are successive ACF peak amplitudes. For lightly damped structures ($\zeta \ll 1$), this simplifies to $\zeta \approx \frac{\delta}{2\pi}$.

To quantitatively evaluate the temporal consistency and robustness of the identified modes, the MAC is employed to calculate the linear correlation between mode shape vectors extracted across different time periods:

$$MAC(\phi_i, \phi_j) = \frac{|\phi_i^H \phi_j|^2}{(\phi_i^H \phi_i)(\phi_j^H \phi_j)} \quad (4)$$

where ϕ_i and ϕ_j are the i -th and j -th strain mode shape vectors extracted from different datasets, and the superscript H denotes the conjugate transpose. The MAC values range from 0 to 1, with values closer to 1 indicating higher similarity.

Results and Discussion

First, to precisely localise the bridge within the

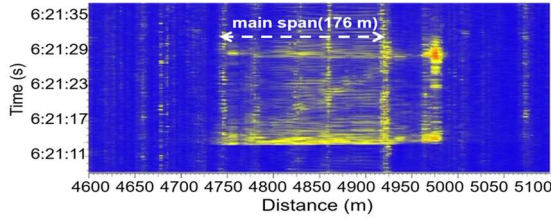


Fig. 2: DAS waterfall diagram for Menai bridge localisation.

57-km telecom fibre, we observed and analysed the waterfall diagram from the DAS. As shown in Fig. 2, a distinct region of intense, coherent vibration was clearly visible, corresponding to the 176 m main span of the Menai suspension bridge. This signature arises from vehicle-bridge interactions and wind-induced vibrations, which produce significantly higher amplitude responses than the surrounding static fibre segments. Moreover, we observed distinct vibration patterns on both sides of the bridge's anchorage area. This is expected as the main supporting cables are anchored differently at each end of the bridge.

The global spectral characteristics of the bridge response are first illustrated in Fig. 3, which presents the PSD heatmap of the bridge main span. Consistent frequency bands were observed across the span, indicating the presence of dominant structural modes with strong spatial coherence.

Modal identification was performed via the EFDD method. As shown in Fig. 4, the first singular value exhibits clear, well-separated peaks at 0.51 Hz and 0.73 Hz (the bridge's first two natural frequencies), while higher-order singular values are significantly lower. This indicates that the ambient structural response is governed by a small number of dominant global modes, which is typical for long-span suspension bridges.

The identified strain mode shapes are shown in Fig. 5. The results exhibit continuous spatial distributions along the bridge span, capturing the structure's global deformation patterns. The dense spatial sampling provided by DAS enables high-resolution modal reconstruction, which would be difficult to achieve using conventional discrete sensing approaches.

The dynamic characteristics of the structure were further validated via damping estimation.

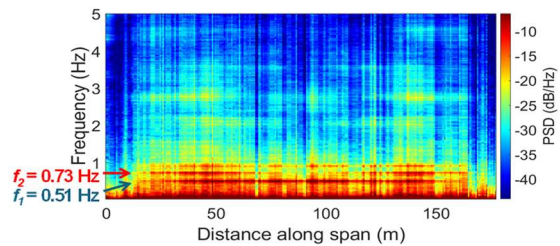


Fig. 3: Power spectral density (PSD) heatmap of the bridge main span, with the first two natural frequencies annotated.

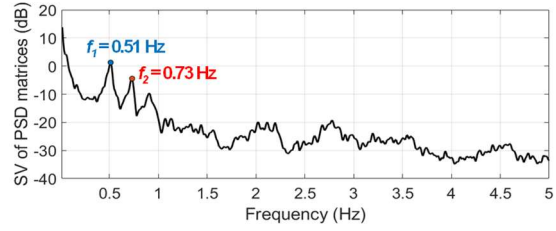


Fig. 4: First singular value of PSD matrices for natural frequency identification.

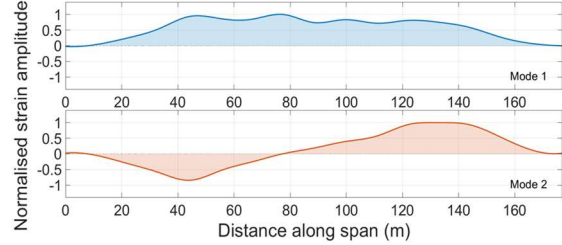


Fig. 5: Normalised strain mode shapes of the first two modes.

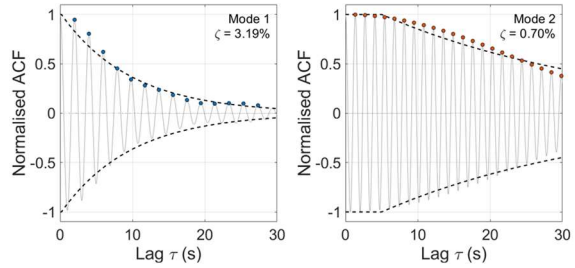


Fig. 6: Normalised autocorrelation functions for modal damping estimation.

Figure 6 illustrates the free-decay ACF and the exponential fit results, indicating damping ratios of 3.19% for Mode 1 and 0.70% for Mode 2. The stable, physically meaningful decay trends confirm that the estimated values fall within the expected range for large civil structures.

Finally, the consistency of the identified modes was evaluated via the MAC, as summarised in Table 1. The diagonal MAC values exceed 0.92, indicating strong repeatability and robustness of the identified modal parameters, while low off-diagonal values confirm good modal orthogonality.

Tab. 1: MAC value matrix of strain mode shapes for estimating modal temporal consistency.

	Mode1	Mode2
	0.927	0.100
Mode1	0.927	0.100
Mode2	0.008	0.933

Conclusions

We have performed the field demonstrations of OSMA for the historic Menai suspension bridge. The bridge health parameters, including natural frequencies, mode shapes, and damping ratios, have been successfully estimated and validated using MAC from data taken at different times.

Acknowledgements

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